Longitudinal Motor Network Responses to Pallidal DBS in Cervical Dystonia: An Ongoing DBS-fMRI Study

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Objective

This longitudinal MRI study in cervical dystonia uses functional MRI (fMRI) to evaluate neural mechanisms underlying clinical response to deep brain stimulation (DBS), with the goal of **identifying imaging biomarkers for early DBS programming**.

Motivation

- DBS treats dystonia effectively, but its **mechanisms remain unclear** and ~20% of patients benefit little[1], with improvement being **slow**
- Prior fMRI work has used single postoperative timepoints[2,3]; **multi-timepoint imaging is needed** to capture pre-to-post effects[4]
- Negative pallidal-sensorimotor, positive pallidal-cerebellar connectivity predict DBS-induced improvement across dystonia subtypes, but has **never been validated longitudinally**

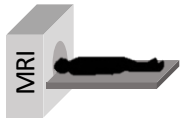
Cohort

N, Patients	Mean age	Sex
12	57.4 ± 11.5	7 F / 5 M
Target	**Preop fMRI**	**Postop fMRI**
11 GPi / 1 VIM	13 (runs)	18 (runs)

Methods

1. FMRI

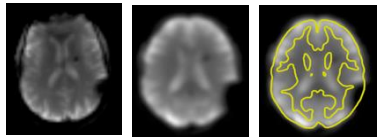
PREOP & POSTOP DBS ON/OFF



DBS device MRI mode*

2. PREPROCESS

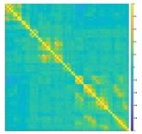
CONN TOOLBOX PREPROCESSING



Raw Smoothed MNI Alignment

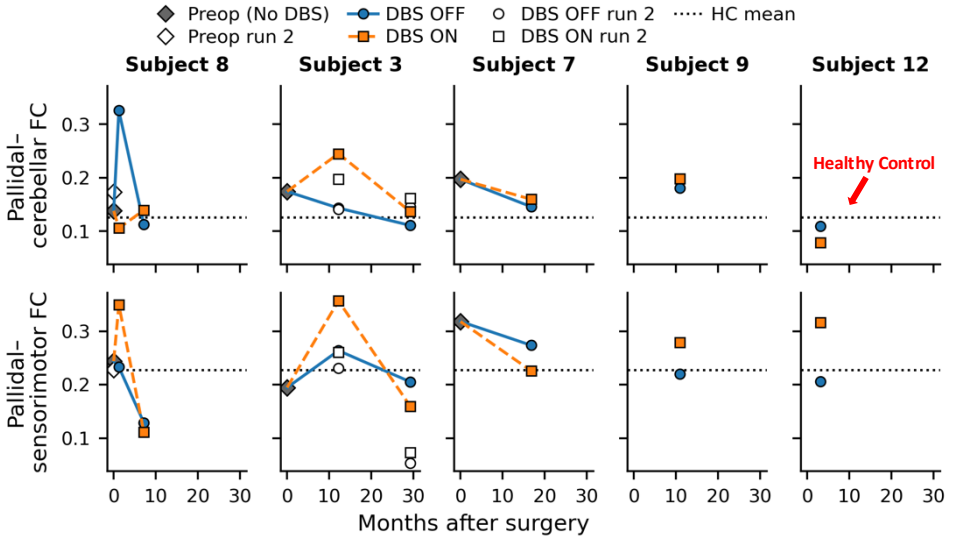
3. EXTRACT METRICS

CONNECTIVITY



Pallidal-cerebellar
Pallidal-sensorimotor
Networks

Results

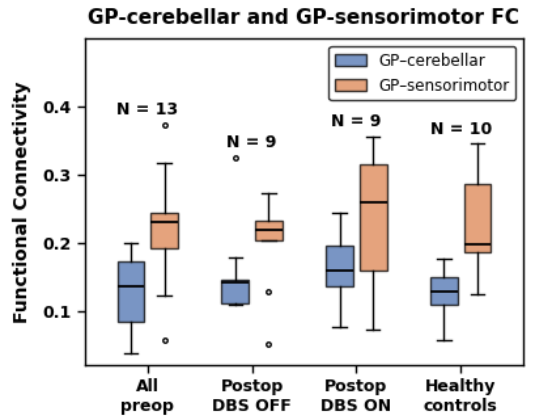


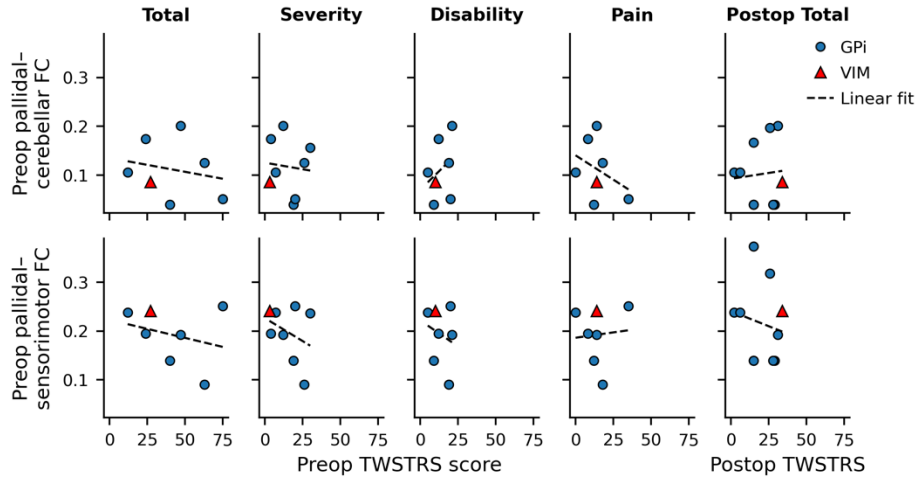
Pallidal-cerebellar and pallidal-sensorimotor connectivity generally declined over time, with transient increases during early optimization. DBS ON vs. OFF differences were most pronounced in the first year; in some patients, connectivity approached control levels.

Continued Results

For all runs included, no group-level trends were observed.

Lower pallidal-cerebellar connectivity trended with worse TWSTRS scores; higher pallidal-sensorimotor connectivity with worse pain only. Baseline connectivity also showed weak relationships with postoperative TWSTRS outcomes.





Discussion

These early results suggest pallidal connectivity is modulated by DBS over time and may contribute to dystonia severity and treatment response, though further investigation is needed.

References